

VETO: Evidence-Constrained Harness Selection for Reliable Chart Reasoning

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Abstract

Optimizing an executable harness for current task accuracy does not explicitly test whether it responds correctly when answer-relevant visual evidence changes. We study paired visual evidence as a safety audit before a subsequent weight update. VETO retains the incoming harness and admits a candidate only when a fixed paired audit supports both marginal-competence non-inferiority and no increase in one-sided fragility. Ambiguous candidates are excluded rather than ranked by a noisy point estimate. The weight update reuses WHALE’s native rejection-sampling fine-tuning. Our prospective study uses a shared scoring protocol, matched candidate archives and independently seeded training branches. **Study status: experiments are incomplete; this draft does not claim an independent method benefit.**

1. Introduction

Chart reasoning requires answers to follow the information in the presented chart. Counterfactual chart evaluation already exposes failures hidden by single-chart scores [3]. Harness optimization and non-regression acceptance also have established precedents [1, 2]. We examine the narrower question of whether paired evidence is useful for choosing the harness that supplies subsequent successful training trajectories.

Our intended contribution is an experimentally supported selection mechanism, not the invention of harness-weight alternation or counterfactual evaluation. The prospective protocol calls for one training–search–training cycle and compares the safety audit against ordinary WHALE selection, a same-data point gate, and counterfactual augmentation. The current V3 study stopped before the second training stage because all three selection rules retained the same incoming harness. No long-horizon causal propagation or population reliability guarantee is assumed.

2. Method

Let θ denote fixed model weights, h_{in} the phase incoming harness, and $C = \{(x_i, x'_i, q_i, y_i, y'_i)\}$ valid answer-changing image pairs. With correctness indicators c_i and c'_i , define the marginal competence, one-side fragility and paired correctness

$$\begin{aligned} M_C(\theta, h) &= (2|C|)^{-1} \sum_i (c_i + c'_i), \\ F_C(\theta, h) &= |C|^{-1} \sum_i [c_i \neq c'_i], \\ P_C(\theta, h) &= |C|^{-1} \sum_i c_i c'_i = M_C - \frac{1}{2} F_C. \end{aligned} \quad (1)$$

where brackets denote the binary indicator. For each candidate, we form source-paired deltas against h_{in} . Let $[L_M, U_M]$ and $[L_F, U_F]$ denote multiplicity-adjusted empirical percentile-bootstrap bounds at the fixed audit size. A candidate is **PASS** if

$$L_M \geq -0.01 \quad \text{and} \quad U_F \leq 0. \quad (2)$$

A candidate is **FAIL** when $U_M < -0.01$ or $L_F > 0$; all remaining candidates are **UNCERTAIN**. Only **PASS** candidates and the always-retained incoming harness form $\mathcal{H}_{\text{ok}}$ and enter the original WHALE ranking:

$$h_{\text{next}} = \arg \max_{h \in \mathcal{H}_{\text{ok}}} (A_H(\theta, h), -\text{turns}_H(\theta, h)). \quad (3)$$

Exact ties retain the original archive order. The paired audit is a constraint, not the ranking objective. A subsequent native RSFT stage uses successful trajectories under h_{next} ; its objective and success filter are unchanged. These empirical constraints do not guarantee independent test accuracy or non-regression after training.

The answer parser and verifier are shared across all conditions and cannot be changed by candidates. Candidate code runs with file and network confinement. Current weights, incoming harness, audit sample identities and decoding settings are fixed within a selection phase.

Equation 1 shows why paired correctness alone is insufficient: it can increase through either higher marginal

VETO: test visual evidence before choosing the next training harness

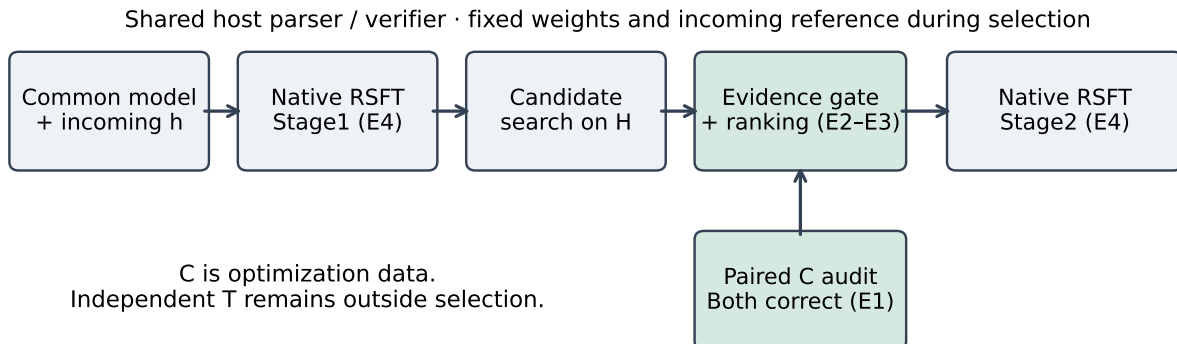


Figure 1. Paired evidence constrains the harness selected for the next native RSFT stage. The shared scorer, candidate search and weight optimizer are common infrastructure.

competence or lower one-side fragility. VETO audits the two components directly, while retaining P_C as the primary downstream endpoint. The bounds are an operational decision rule on the fixed optimization set; they are not a distribution-free confidence sequence or a population guarantee. Joint correctness does not identify why an answer fails. Perception, arithmetic, instruction following and answer formatting can all affect this endpoint; paired failures alone do not establish visual non-use or a latent mechanism.

3. Experimental Protocol

We use the public post-trained Qwen3.5-4B checkpoint [6] and seeds 42, 43, 44. Each weight stage has four batches, eight questions per batch and eight trajectories per question. Each search attempts at most three candidates. Identical first stages and candidate measurements are shared within a seed; subsequent branches are evaluated as dependent, matched comparisons rather than independent first-stage trials.

PlotQA-EvidencePairs is a separately named dataset derived from PlotQA source tables [5], with comparison, value-reading and difference questions. All variants of a source table remain in the same partition: W2048, H128, V256 pairs, T1024 pairs, and an independent recheck pool. V3 additionally uses answer-changing twins of the 128 H sources as proposal feedback and a fresh C256 audit drawn from source tables outside every historical partition. C is optimization data. Numeric answers use a declared tolerance; answer-changing intervals must not overlap. The external ChartQA [4] evaluation uses a preregistered subset of 256 human and 256 augmented questions; it is not the full official benchmark score.

The common initial prompt and learning rate are selected

Condition	Single image	Both images
Weight only	—	—
Harness only	—	—
WHALE, one cycle	—	—
VETO, one cycle	—	—

Table 1. Template only. No formal condition has a completed scientific result. Final tables will include all seeds, mean, sample standard deviation and source-cluster confidence intervals.

using ordinary development accuracy only. The final checkpoint is determined by the frozen budget. Test data are excluded from all model and harness selection.

4. Results

At fixed seed 42 post-stage-one weights, the complete V3 optimization archive scored h0–h3 at 108, 106, 63, 107 correct on H128. Their two-image correctness on fresh C256 was 203, 186, 93, 205 pairs, with 439, 414, 240, 442 correct individual images. The safety audit classified h1 and h2 as **FAIL** and h3 as **UNCERTAIN**. WHALE, VETO, and the matched point-estimate gate therefore all selected h0. This failed the preregistered distinct-selection condition, so we did not run V3 continuation branches. These optimization-set measurements do not establish a post-training VETO benefit.

The same-data marginal gate and counterfactual augmentation are mandatory controls. If gates select the same harness, that observation provides no evidence for a distinct benefit of VETO. We check the error decomposition on the actual fixed-weight C archives; calibration measurements at different weights cannot establish that the acceptance rules

Selection / control	Paired score	Difference
Ordinary accuracy	—	—
Same-C marginal gate	—	—
VETO safety audit	—	—
50% paired augmentation	—	—

Table 2. Mandatory controls. V3 continuation was not executed because the three selection rules chose the same incoming harness.

Condition	Human	Augmented	Cost
WHALE	—	—	—
VETO	—	—	—
Controls	—	—	—

Table 3. ChartQA fixed subset and total GPU/API costs. No external model score is available in this scaffold.

- tion Answering about Charts with Visual and Logical Reasoning. Findings of ACL,2022. 2
- [5] Nitesh Methani, Pritha Ganguly, Mitesh M. Khapra, and Pratyush Kumar. PlotQA: Reasoning over Scientific Plots. WACV,2020. 2
- [6] Qwen Team. Qwen3.5: Towards Native Multimodal Agents.2026. qwen.ai/blog?id=qwen3.5. 2

choose different harnesses.

Stage trajectories and performance–cost plots will be generated from complete branch records. Empty plots or simulated improvements are not experimental evidence.

5. Limitations and Decision Criteria

The study is limited to one model size, one domain and one complete alternation. Source-derived renderings do not establish general visual-agent reliability. The native checkpoint protocol omits Adam moments; all branches use the same restoration behavior. We report full allocation costs including failed runs, loading, candidate evaluation and audit overhead.

The prospective candidate criterion requires at least one percentage point mean paired improvement, nonnegative differences across all three seeds, positive differences in at least two seeds, ordinary accuracy loss no greater than one point, and a positive lower bound of the paired source-cluster95% interval. Independent value beyond a marginal gate must also be demonstrated. Unsupported outcomes will be reported as inconclusive rather than hidden.

References

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